

Midterm Potential Topics 10/09 5 - 7PM

Module 1: Networking Basics

- What is networking and its primary purposes
 - Types of networks (PAN, LAN, WAN, WLAN, CAN, MAN, SAN, VPN, etc.)
 - IP addresses – IPv4 vs IPv6, static vs dynamic
 - TCP/IP protocol suite and its four layers (Link, Internet, Transport, Application)
 - Function and structure of packets (header, payload, trailer)
 - Roles of routers and routing tables
 - TCP vs UDP and the concept of connection-oriented vs connectionless communication
 - TCP three-way handshake (SYN, SYN-ACK, ACK) and termination (FIN, ACK)
 - Common port numbers and associated protocols (HTTP, HTTPS, SSH, DNS, SMTP, etc.)
 - MAC address and NIC cards
 - Packet sniffing basics and tools (Wireshark, tcpdump, scapy)
 - Packet spoofing, DDoS, and man-in-the-middle (MiTM) attacks
 - Packet filtering (ingress vs egress) and protection methods against spoofing
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Module 2: Cyber Kill Chain

- Purpose and origin of the Cyber Kill Chain (Lockheed Martin)
- The 7 stages of the Cyber Kill Chain:
 1. Reconnaissance
 2. Weaponization
 3. Delivery
 4. Exploitation
 5. Installation
 6. Command & Control (C2)
 7. Actions on Objectives
- Examples of each phase (Target retail breach case study)
- How each phase can be disrupted or mitigated
- Role of phishing and social engineering in Delivery phase

- Weaknesses of the Cyber Kill Chain (e.g., lack of insider threat detection, static model)
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Module 3: Reconnaissance

- Purpose and importance of reconnaissance in ethical hacking
 - Difference between passive and active reconnaissance
 - Legal implications of active reconnaissance
 - Examples and how to use passive reconnaissance tools:
 - WHOIS, nslookup, dig
 - Examples and how to use active reconnaissance tools:
 - Ping, traceroute, telnet
 - Information collected during recon (network, host, employee, policy data)
 - How attackers and ethical hackers use recon data to identify vulnerabilities
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Module 4: Nmap

- What Nmap is and its purpose in network scanning
 - Legal implications of using Nmap
 - Basic Nmap syntax and scanning techniques:
 - Port scanning: `nmap -p , --top-ports , -iL`
 - Saving results: `-oN`
 - Service and OS detection: `-A , -sV` , and timing templates (`-T1 – -T5`)
 - TCP vs UDP scanning (`-sT` and `-sU`)
 - Nmap Scripting Engine (NSE):
 - Purpose and capabilities
 - Example scripts (`--script vuln , http-malware-host`)
 - Reading and interpreting Nmap results (open, closed, filtered ports)
 - Common real-world applications of Nmap in penetration testing
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Module 5: Firewalls – Potential Topics

- What a firewall is and how it protects a network
- Hardware vs software firewalls (host-based vs network-based)

- Blacklist, whitelist, and greylist – definitions, pros, and cons
- MAC address filtering and its purpose
- Access Control Lists (ACLs) – how they control network traffic
- Stateless vs stateful firewalls – differences and examples
- Web Application Firewall (WAF) – how it protects web apps
- Next-Generation Firewall (NGFW) – deep packet inspection, content filtering
- Configuring basic firewall rules (allow/deny by IP, port, or protocol)
- Common firewall use cases in enterprise and home networks